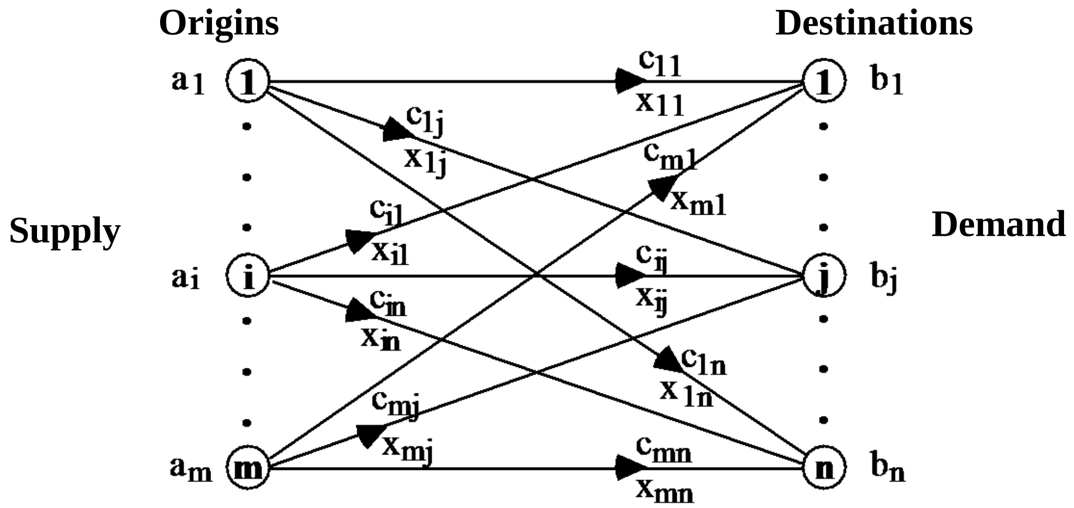
V – Particular Problems of Linear Programming

Transportation problem

It admits a lot of variants but, in its simples form, it consists in determining the most economical way of sending a product that is available in limited quantities in certain places (origins) to other places where it is necessary (destinations).



Therefore, it consists in determining the optimal plan for <u>transporting a product</u> from <u>m origins</u> to <u>n destinations</u>.

Where:

- x_{ij} = Quantity of product to transport from origin i to destination j
- c_{ij} = Cost of transporting one unit of product from origin i to destination j
- a_i = Quantity of product available in origin i
- b_j = Quantity of product required in destination j

The table below facilitates the formalization of a transportation problem:

Destinations / **Origins**	1	2	...	n	**Supply**
1	c_{11} x_{11}	c_{12} x_{12}	...	c_{1n} x_{1n}	a_1
2	c_{21} x_{21}	c_{22} x_{22}	...	c_{2n} x_{2n}	a_2
...	...	...	...	...	...
m	c_{m1} x_{m1}	c_{m2} x_{m2}	...	c_{mn} x_{mn}	a_m
Demand	b_1	b_2	...	b_n	$\sum a_i = \sum b_j$

The mathematical formalization of a transportation problem as a linear programming problem:

Determining x_{ij} = quantity of product to transport form origin i to destination j in order to minimize the total transportation cost, that is,

$$\text{Min } z = c_{11}x_{11}+...+c_{1n}x_{1n}+... ...+c_{m1}x_{m1}+...+c_{mn}x_{mn}$$

subject to

$$\left.\begin{array}{l} x_{11}+x_{12}+...+x_{1(n-1)}+x_{1n} = a_1 \\ ... \qquad ... \qquad ... \\ x_{m1}+x_{m2}+...+x_{m(n-1)}+x_{mn} = a_m \end{array}\right\} \textbf{\textit{Supply}}$$

$$\left.\begin{array}{l} x_{11}+x_{21}+...+x_{(m-1)1}+x_{m1} = b_1 \\ ... \qquad ... \qquad ... \\ x_{1n}+x_{2n}+...+x_{(m-1)n}+x_{mn} = b_n \end{array}\right\} \textbf{\textit{Demand}}$$

$$x_{ij} \geq 0;\ i = 1, 2, ..., m;\ j = 1, 2, ..., n$$

that is

Determining

$$x_{ij}$$

in order to minimize

$$z = \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij} x_{ij}$$

subject to

$$\sum_{j=1}^{n} x_{ij} = a_i \text{ (\textbf{\textit{supply constraints}})}$$

$$\sum_{i=1}^{m} x_{ij} = b_j \text{ (\textbf{\textit{demand constraints}})}$$

$$x_{ij} \geq 0;\ i = 1, 2, ..., m;\ j = 1, 2, ..., n$$

Example

A company has 3 warehouses (**O1**, **O2** and **O3**) where there are available **7**, **10** and **12** units of a given product, respectively, which are transported to 4 markets (**D1**, **D2**, **D3** and **D4**) that consume **4**, **8**, **11** and **6** units, respectively.

The unitary transportation costs (expressed in M.U. – monetary units), from the several origins to the various destinations, are the ones presented in the following table:

	D_1	D_2	D_3	D_4
O_1	12	3	8	4
O_2	7	4	6	9
O_3	8	7	3	6

Mathematical formalization of the problem:

Determine

x_{ij} = quantity of product to transport from warehouse Oi to the market Dj (i = 1, 2, 3; j = 1, 2, 3, 4)

in order to minimize

$$z = 12\,x_{11} + 3\,x_{12} + 8\,x_{13} + 4\,x_{14} + 7\,x_{21} + 4\,x_{22} + 6\,x_{23} + 9\,x_{24} + 8\,x_{31} + 7\,x_{32} + 3\,x_{33} + 6\,x_{34}$$

subject to

$$
\begin{array}{llllllllllllll}
x_{11}+x_{12}+x_{13}+x_{14} & & & & & & & & & & & & = & 7 \\
& x_{21}+x_{22}+x_{23}+x_{24} & & & & & & & & & & & = & 10 \\
& & x_{31}+x_{32}+x_{33}+x_{34} & & & & & & & & & & = & 12 \\
x_{11} & +x_{21} & +x_{31} & & & & & & & & & & = & 4 \\
x_{12} & +x_{22} & +x_{32} & & & & & & & & & & = & 8 \\
x_{13} & +x_{23} & +x_{33} & & & & & & & & & & = & 11 \\
x_{14} & +x_{24} & +x_{34} & & & & & & & & & & = & 6
\end{array}
$$

$x_{ij} \geq 0$; $i = 1, 2, 3$; $j = 1, 2, 3, 4$

A problem of this type can be solved by the Simplex method:

m+n *constraints*

$m*n + \underbrace{(m+n)}_{\textit{artificial}}$ *variables*

However, it is complicated!

In effect, due to its particular structure, these problems can be solved by a <u>special algorithm</u>.

Some important properties of transportation problems

- The transportation problem has always a finite optimal solution.

- Any feasible basic solution of a transportation problem has, at the maximum, m+n-1 positive variables.

  In fact, in this kind of problems there always exist m+n-1 basic variables since one of the m+n constraints is redundant and, thus, can be obtained as a linear combination of the remain m+n-1 constraints. Since in case of a degenerated solution some of the basic variables can be zero, there exist at the maximum, m+n-1 positive variables.

- If a_i and b_j ($i = 1, 2, ..., m$; $j = 1, 2 ..., n$) are integer, then any feasible basic solution has only integer values.

Transportation problem resolution

The methods used to solve transportation problems are “variants” of Simplex method and are based on the particular structure of this kind of problems.

They involve the following steps

1 - Determination of the initial FBS

2 - Test of the optimal:

- If the current FBS satisfies the test ➔ end
- Otherwise ➔ proceeds

3 - Improvement of the solution:

- Calculation of the new FBS through the introduction in the basis of a non basic variable replacing one basic variable
- Go back to 2

1 - For obtaining the first FBS the following methods can be used:

- **Northwest Corner**
- **Matrix Minimum**
- **Vogel’s Approximation**

Example

Consider the previous example (page VI-5).

Using the tabular representation the following table is obtained:

	D1	D2	D3	D4	
O1	12 x11	3 x12	8 x13	4 x14	7
O2	7 x21	4 x22	6 x23	9 x24	10
O3	8 x31	7 x32	3 x33	6 x34	12
	4	8	11	6	29 = 29

Northwest Corner

The basic variable will be, in each iteration, the one that is localized in the superior left corner. It is fulfilled with the minimum of the values of the corresponding "ai" and "bj" (that is, the maximum as possible). Then, the row or the column in which the availability or the requirement became zero is "eliminated".

Repeat the process

For the same example:

	D1	D2	D3	D4	
O1	12 4(1º)	3 3(2º)	8 -------------	4 -------------	$\cancel{7}$ $\cancel{3}$
O2	7 |	4 5(3º)	6 5(4º)	9 -------------	$\cancel{10}$ $\cancel{5}$
O3	8 |	7 |	3 6(5º)	6 6(6º)	$\cancel{12}$ $\cancel{6}$
	$\cancel{4}$	$\cancel{8}$ $\cancel{5}$	$\cancel{11}$ $\cancel{6}$	$\cancel{6}$	

A possible solution was obtained.
Simple method but that does not consider the costs.

Cost = 4*12 + 3*3 + 5*4 + 5*6 + 6*3 + 6*6 = 161

Matrix Minimum

The basic variable will be, in each iteration, the one that has the least cost of the table. It is fulfilled with the minimum of the values of the corresponding "ai" and "bj" (that is, the maximum as possible). Then, the row or the column in which the availability or the requirement became zero is “eliminated”.

Repeat the process

Considering the same example:

	D1	D2	D3	D4	
O1	12 -------------	3 7(1º, 2º)	8 -------------	4 -------------	~~7~~
O2	7 4(5º)	4 1(3º)	6 |	9 5(6º)	~~10~~ ~~9~~ ~~5~~
O3	8 -------------	7 |	3 11(1º, 2º)	6 1(4º)	~~12~~ ~~1~~
	~~4~~	~~8~~ ~~1~~	~~11~~	~~6~~ ~~5~~	

Another possible solution is obtained (closer to the optimal one).

Cost = 7*3 + 4*7 + 1*4 + 5*9 +11*3 + 1*6 =

= 137 < 161 (previous)

Vogel’s Approximation

The penalties of the rows and columns of the table are initially calculated. A penalty is the difference between the two lowest transportation costs of the cells (not yet “eliminated”) corresponding to the row or column under consideration. In the row or column with the biggest penalty, the cell with the lowest cost is chosen and fulfilled with the maximum as possible. Then, the row or the column in which the availability or the requirement became zero is “eliminated”.

Repeat the process

Considering the same example:

	D1	D2	D3	D4	
O1	12 1(4º,5º,6º)	3 |	8 |	4 6(3º)	~~7~~ ~~1~~
O2	7 2(4º,5º,6º)	4 8(2º)	6 |	9 |	~~10~~ ~~2~~
O3	8 1(4º,5º,6º)	7 |	3 11(1º)	6 |	~~12~~ ~~1~~
	~~4~~ ~~3~~ ~~1~~	~~8~~	~~11~~	~~6~~	

Another possible solution is obtained (better than the previous ones).

Cost = 12*1 + 2*7 + 8*1 + 8*4 + 11*3 + 6*4 = 123 < 137 < 161 (previous)

Penalties:

1st Table:

1st row = 1; 2nd row = 2; 3rd row = ***3***

1st column = 1; 2nd column = 1; 3rd column = ***3***; 4rd column = 2

↑

for example

2nd Table:

1st row = 1; 2nd row = ***3***; 3rd row = 1

1st column = 1; 2nd column = 1; 4rd column = 2

3rd Table:

1st row = ***8***; 2nd row = 2; 3rd row = 2

1st column = 1; 4rd column = 2